

DIGITAL PUBLIC GOOD · AGRICULTURE

AgriN

Field-specific farm advice for Indian smallholders, from satellite, weather and ISRO soil data — with crop-disease diagnosis that runs on the phone itself.

Every number is computed. Gemini only explains it.

Advice exists. It just isn't about your field.

Advisories are issued per district. A district is thousands of square kilometres and hundreds of soil types. Two farmers three kilometres apart, on sand and on clay, get the same irrigation advice. One of them is wrong.

The advice arrives as a broadcast — a bulletin, an SMS blast — with no way to ask "what about mine?"

And the new AI farm assistants hallucinate doses. A chatbot that invents "apply 120 kg of urea" is not a smaller version of good advice. It is a crop-failure liability with a friendly tone.

The gap we target

Per-field, computed, explainable — and in a language the farmer actually reads or hears.

The 86% who farm under two hectares

< 2 ha

Small and marginal holdings — the overwhelming majority of Indian farms, and the least served by extension services.

1 : 1,000s

Extension officer to farmer ratios make per-field human advice impossible at scale.

6

Languages with written *and spoken* advisories today: English, Hindi, Punjabi, Bengali, Marathi, Telugu.

Design consequence, not a footnote: agricultural literacy skews lower, older and more female than the national average. An app that only writes excludes the people it is for. Every advisory can be listened to.

Nothing in the product assumes a smartphone-literate user, a reliable connection, or the ability to read a paragraph. On-device diagnosis works with the network off; the deterministic template works with no API key at all.

One loop, end to end, working today

Map the field	Set crop & sowing	Node ingests	Engine computes	Gemini explains
Walk the boundary, or draw it on satellite imagery	14 crops, FAO-56 coefficients	Sentinel-2 NDVI, weather, ISRO soil	Water balance, N-P-K, crop health	In the farmer's language, read aloud

Plus disease diagnosis from a leaf photograph, which needs no field mapped at all — a 1.14 MB network runs on the handset, with no signal and no data cost.

- FAO-56 Penman–Monteith reference evapotranspiration
- Soil-water balance, with a separate ponded model and IRRI safe AWD for paddy
- Nitrogen, phosphorus, potassium from crop removal and soil fertility class
- Crop health: NDVI against the expected curve for that crop and growth stage

Verified end to end on a real field

Punjab, 0.97 ha: **205 days of weather** and **72 Sentinel-2 observations** ingested, 26 intervals correctly rejected for cloud, advisory rendered and narrated.

724 backend tests, 106 mobile tests. Both FAO-56 chapter 4 worked examples reproduced exactly.

The model never produces a number

Every recommended quantity is computed deterministically. Gemini only translates and explains those numbers — it never produces them.

This is not a prompt instruction. `ai/guard.py` extracts every figure the model wrote and every figure the engine computed, and **rejects the narration if the model produced one the engine did not.**

Rejection triggers one corrective round naming the offending figures. Fail twice and the farmer sees the deterministic template — the same advice, plainer words.

The same principle governs chemicals: pesticide doses and pre-harvest intervals are served only from rows verified against the national register. That table ships empty, so an unverified node gives cultural and preventive advice only. A model can never introduce a product name.

Proof it belongs to the system, not the vendor

The cloud model was swapped from one provider to Gemini in an afternoon. **guard.py was not touched.** Not one line.

If the safety property had lived in the vendor integration, replacing the vendor would have broken it.

Where Google AI does the work

SERVICE	JOB	WHY IT IS THE RIGHT TOOL
Gemini 3.6 Flash	Advisory narration	Rephrases computed figures into plain language, natively multilingual. Guard-checked.
Gemini 3.6 Flash	Escalated disease diagnosis	Multimodal. Sees only the photographs the on-device model refused — the hard half.
Cloud Text-to-Speech	Spoken advisories	Six Indian locales. Synthesised from guard-passed text only.
Cloud Translation	Template fallback	Translates the deterministic wording, rejecting any translation that moves a digit.
Maps Platform	Satellite basemap	Street tiles give a farmer nothing to trace; imagery shows the actual field edge.
Vertex AI	Same CNN, served	For handsets whose TFLite delegate will not load — not a replacement for on-device.
Earth Engine · BigQuery · Cloud Run	Scale path	Second NDVI supply past the Copernicus quota; national aggregate queries; ministry-native hosting.

Each fails closed and is optional: a node with no key still computes the same irrigation depth and renders it through the deterministic template. The engine has no dependency on any model — which is what makes the guarantee on slide 5 credible rather than aspirational.

It abstains more often than it answers

0.981

Accuracy of predictions **actually shown to a farmer** — those clearing the on-device confidence gate.

44.4%

Of photographs answered on the handset. The rest escalate to Gemini or return inconclusive.

1.14 MB

Model size. Runs offline, on a low-end Android, at no data cost.

The gate is the product. Raw model accuracy is 0.786. Confidence, margin and entropy thresholds convert that into 0.981 on what reaches a farmer, by refusing the rest.

A wrong confident answer costs a season. "Unknown" costs a second photo.

What we will not claim

We hold a 0.994 figure on a 1,677-image test slice. It is the **same collection** as part of the training data and covers 4 of 10 classes. Our own model card says it **must not be quoted** as a cross-collection result — so it isn't, here or anywhere.

Live sources — and one that mattered more than expected

LAYER	SOURCE
Satellite	Copernicus Sentinel-2 L2A, SCL cloud-masked
Weather	Open-Meteo → NASA POWER → MET Norway
Soil	Soil Health Card → farmer feel-test → SoilGrids → ISRO Bhuvan
Agronomy	FAO-56 tables 11, 12, 19, 22

Every figure carries its provenance to the farmer's screen — including when it is a regional estimate rather than a measurement of their own field.

SoilGrids returns null across all of India

Verified against a Brazilian control, which answers normally. So the node fell back to assuming **loam** for every field in the country.

ISRO's Bhuvan national soil map says the test field is **loamy sand**.
FAO-56 Table 19: loam holds 130 mm of available water per metre; loamy sand holds 80.

The node had been overstating that field's water reservoir by more than half — directly into the irrigation decision.

Fallback chains, never blends. A figure is always from one named source, so an error can be traced to it rather than averaged into invisibility.

One node per state. Same code, sovereign data.

Nodes exchange models and aggregate statistics. Never raw farmer data.

- District-level aggregates only, with **k-anonymity and complementary suppression** enforced structurally
- Every published payload **Ed25519-signed** over canonical JSON, so it cannot be replayed under another node's name
- A publish is **aborted** if an identifier is detected in the payload — checked again on the BigQuery export path
- EXIF and GPS stripped from every uploaded photograph

A state department runs its own node, holds its own farmers' data, and still contributes to — and benefits from — a national picture.

Why this shape, not a central platform

A single national database of every farmer's field boundaries and yields is a target and a governance problem. Federation means **no one has to hold it**.

It is also the only shape that survives the politics: states adopt what they control.

Standards are boring on purpose — AGROVOC crop concepts, UCUM units, GeoJSON RFC 7946, OpenAPI 3.1 — so a second implementation can interoperate without asking us.

Live, not planned. The running node declares itself IN-MH at `/well-known/agrin-node`, and serves the real 1.1 MB classifier with its SHA-256. Edit the state out of a signed aggregate and the signature fails.

It is deployed, and it grades itself

A live node runs at `agrin-node-in.onrender.com`. The same container deploys to **Cloud Run** unchanged, for a ministry that already lives inside Google Cloud.

`GET /ready` is the node's own honest self-assessment. Every check names the *user-visible consequence*, not an internal state, and is graded **BLOCKER** **DEGRADED** **OK**.

It currently reports **two blockers**: no SMS provider, so no farmer can sign in yet — and the fixed demo code that let *you* sign in, which it refuses to call acceptable now that the node holds someone else's data. It says both in public, rather than waiting to be found out.

Weeks, not quarters — what a pilot needs

SMS aggregator (DLT-registered)	1–3 weeks
Pesticide rows verified by an agronomist	days
Persistent photo storage	hours
District mapping for aggregates	hours

None is research. All are procurement or configuration — which is why `docs/PIL0T.md` scopes a 15–25 farmer pilot in one district with a working extension officer.

What is not finished

A prototype that hides its gaps is harder to deploy, not easier. Assessed against all nine DPG Alliance indicators — including the two it does not meet.

BLOCKER

SMS delivery. No farmer can sign in until a DLT-registered aggregator is configured. Deliberate: we will not enrol a farmer on a login we cannot deliver.

PARTIAL

Platform independence. The node now calls Google. Every service fails closed and the engine needs none of them — but keyless, the farmer loses plain language in their own tongue. We downgraded our own DPG score rather than argue.

PARTIAL

Independent review. No external agronomic review of the recommendation logic, and no DPIA. Both need people, not code. Neither is bluffed.

Also stated plainly in the repo: the disease model is rice-only, so the other 13 crops route to Gemini and receive a name with no treatment attached; the Earth Engine adapter has not been verified against the live service; and Open-Meteo's free tier is non-commercial, so a paid deployment needs their commercial tier or a self-hosted instance.

Water and nitrogen, which is where the leverage is

Punjab's groundwater is falling. Nitrogen is over-applied across the Indo-Gangetic plain. Both are **decisions taken field by field**, weekly, by millions of people working from a district bulletin.

AgriN does not ask a farmer to trust a model. It tells them the number, where it came from, and how confident it is — including when the answer is "we don't know".

- **SDG 2.3 / 2.4** — smallholder productivity and resilience
- **SDG 6.4** — water-use efficiency, via the AWD paddy model
- **SDG 13** — reduced methane from AWD and reduced nitrogen loss

Next 90 days

Weeks 1–3. SMS aggregator, persistent storage, agronomist-verified pesticide rows for one state.

Weeks 4–10. 15–25 farmer pilot in one district with an extension officer, measuring whether the advice is *followed* — not whether the app works.

Weeks 11–13. Second crop model trained on pilot corrections; second node stood up to prove federation between two operators.